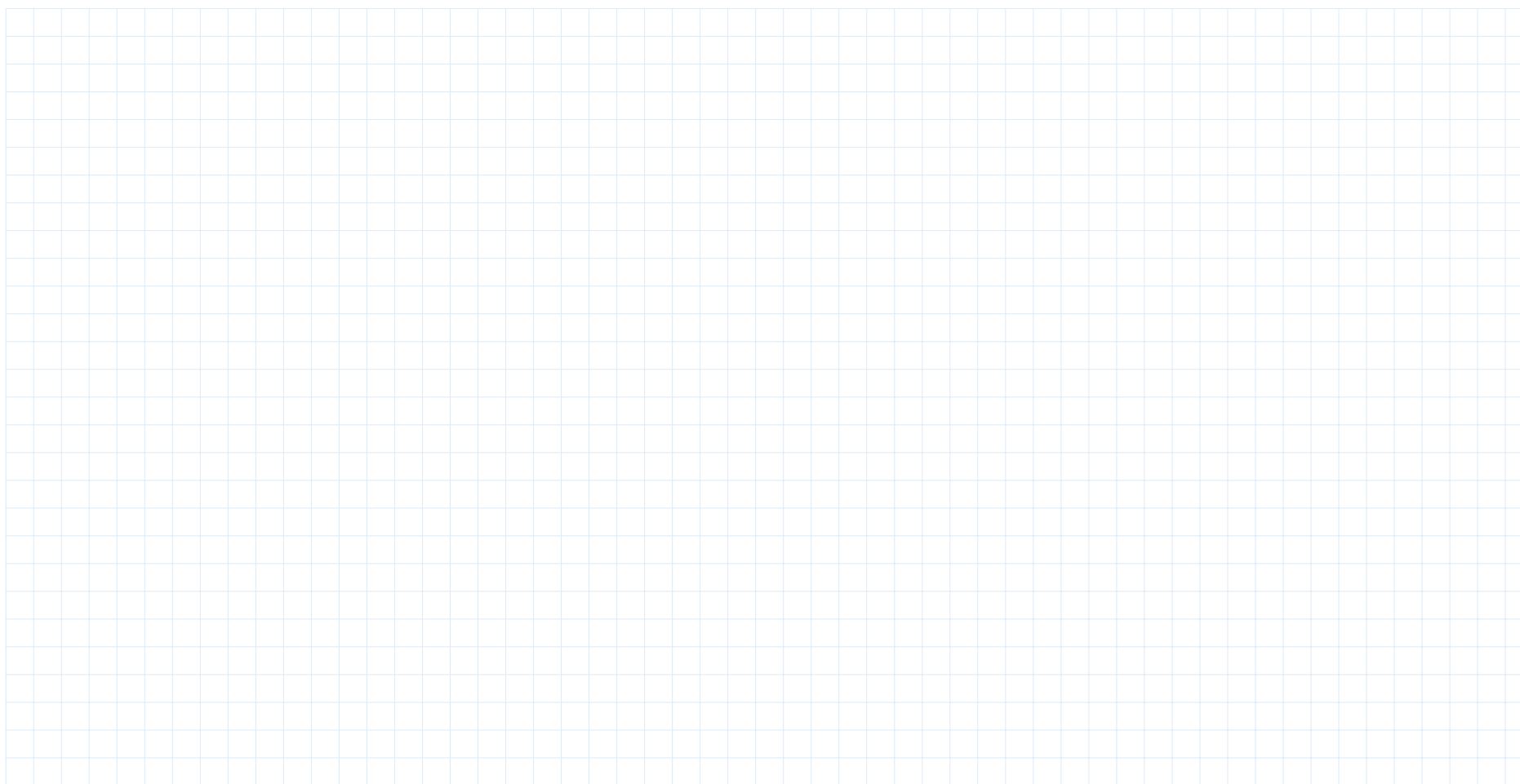


A Triangle for a Composition

If $\theta = \arctan x$, then $\tan \theta = x$ and $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Draw a right triangle and find the exact value:

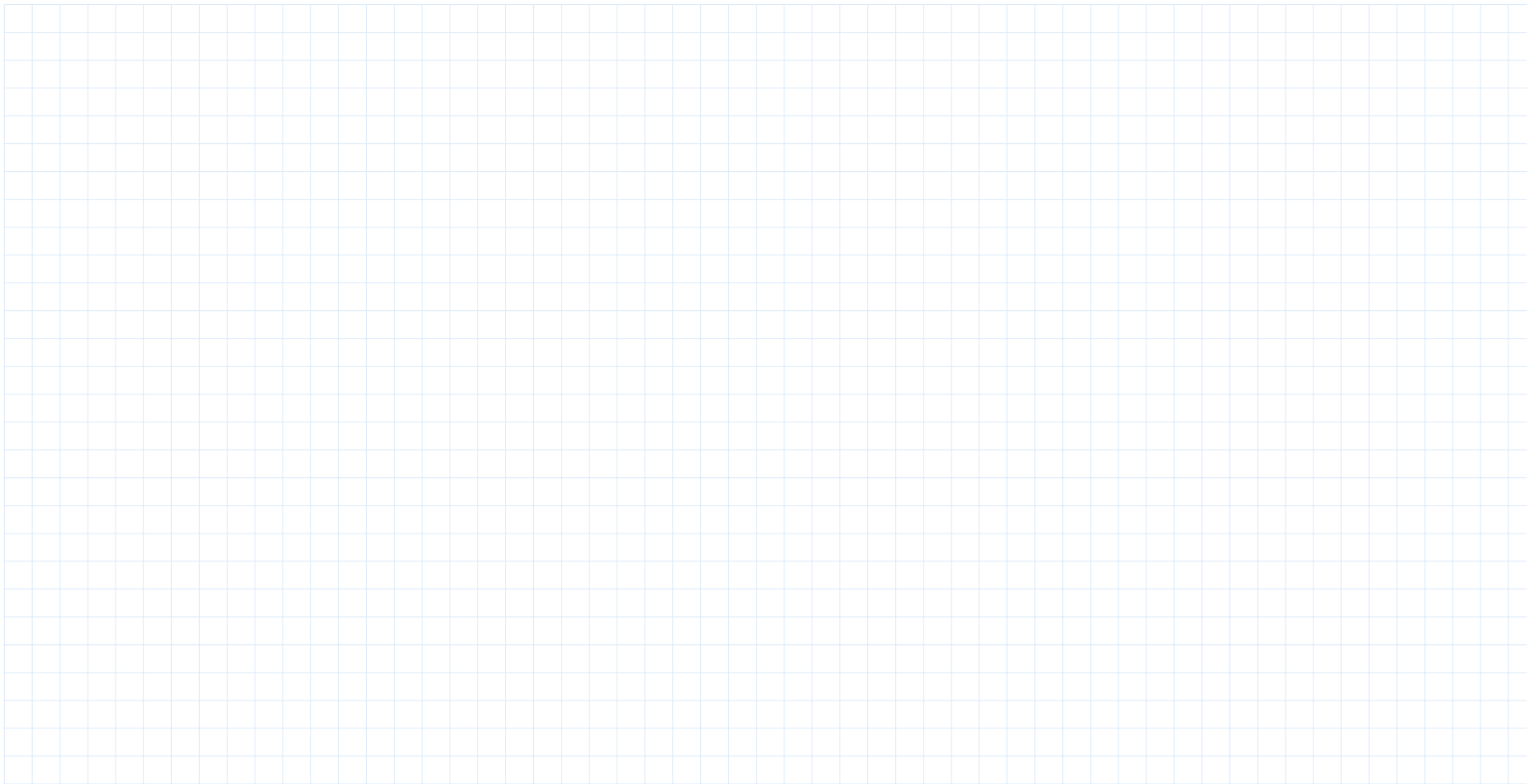
$$\sin\left(\arctan \frac{3}{4}\right).$$



The Quadrant Determines the Sign

Find the exact value. Identify the quadrant before choosing a sign.

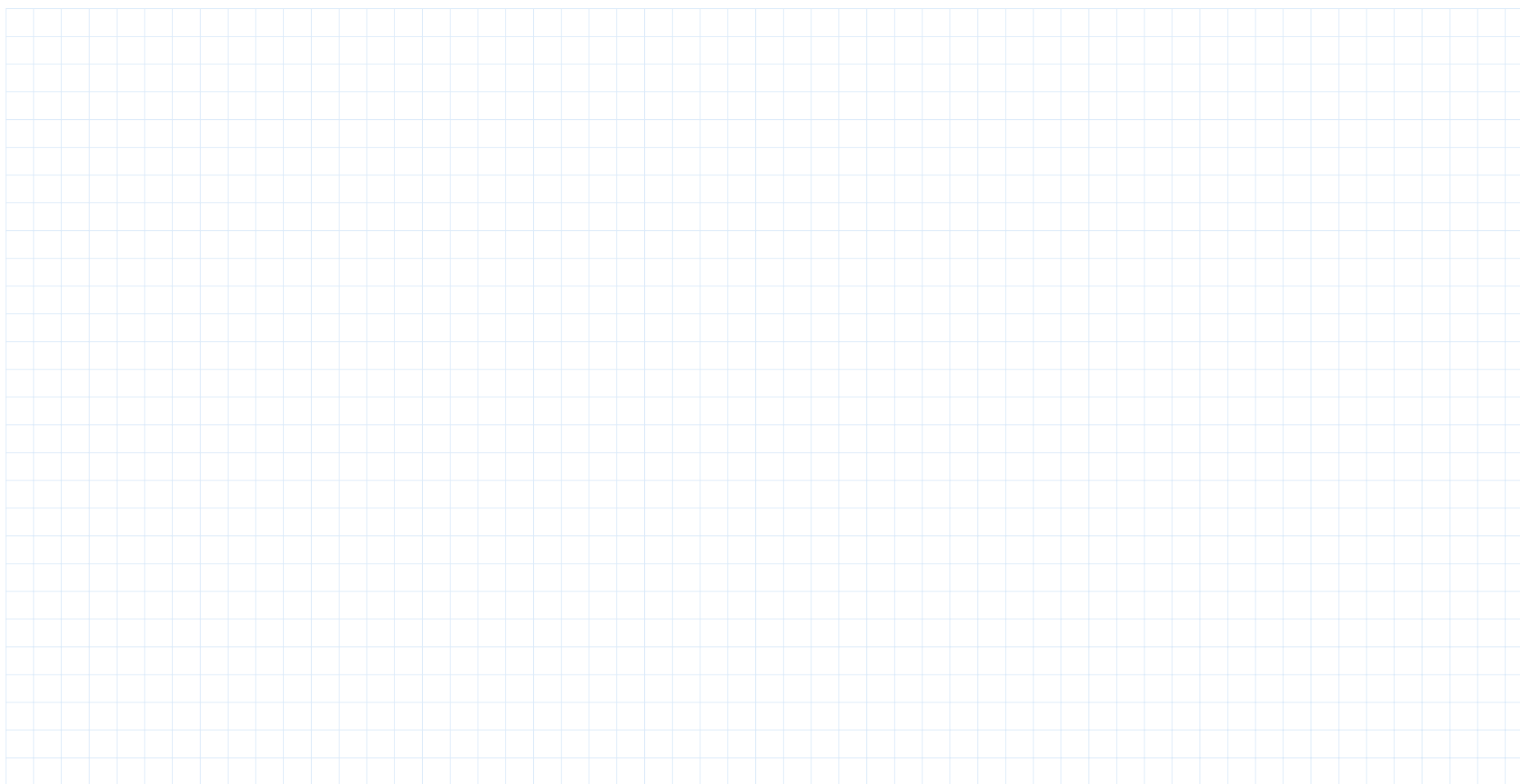
$$\tan\left(\arccos\left(-\frac{5}{13}\right)\right).$$



Inverse Secant and Cosecant

Section 6.2

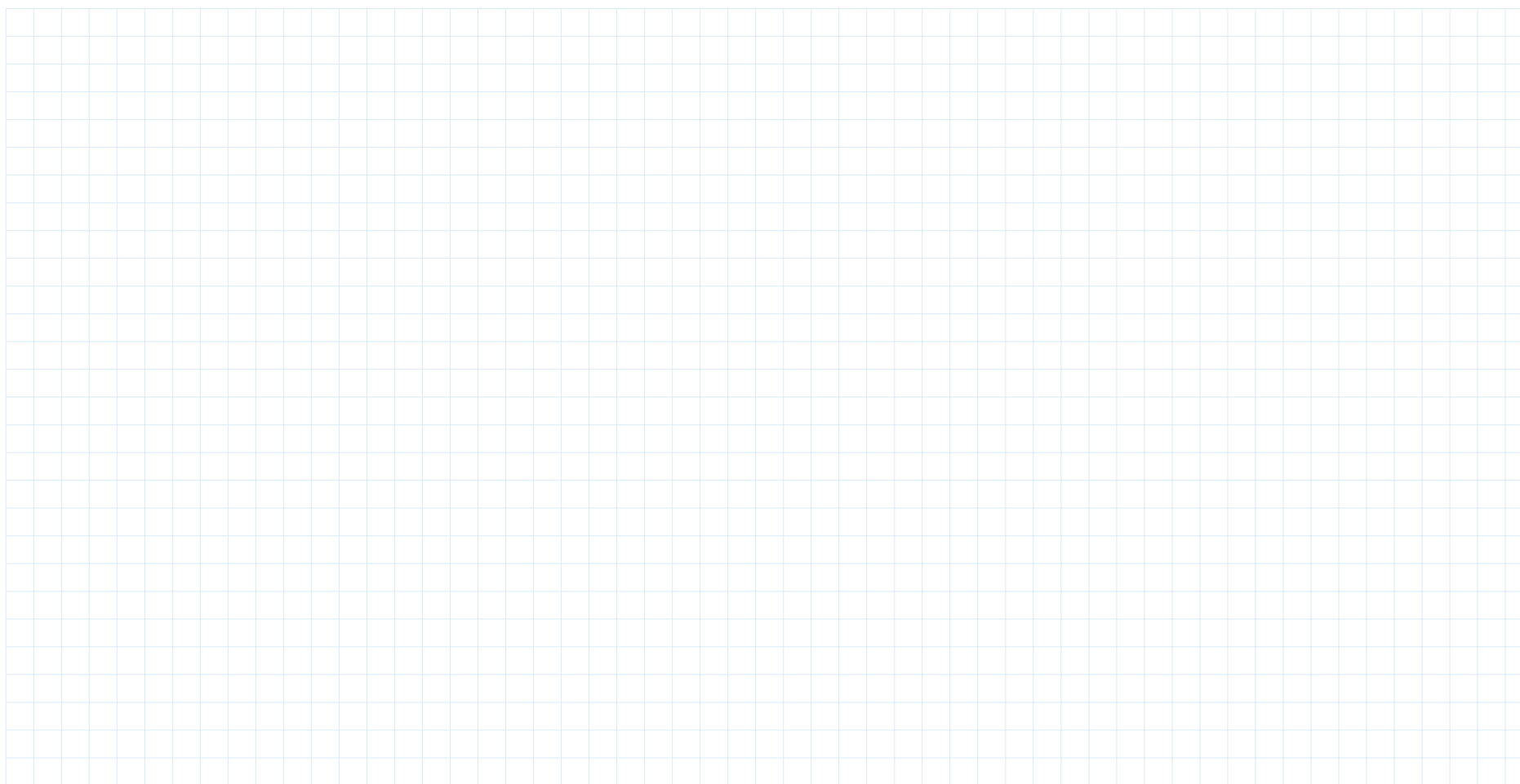
$\operatorname{arcsec} x = \arccos\left(\frac{1}{x}\right)$ and $\operatorname{arccsc} x = \arcsin\left(\frac{1}{x}\right)$, with $|x| \geq 1$. Their ranges are $[0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$ and $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$, respectively. Find the exact values of $\operatorname{arcsec}(-2)$ and $\operatorname{arccsc}(-2)$.



Inverse Cotangent

Section 6.2

$\operatorname{arccot} x$ is the angle in $(0, \pi)$ whose cotangent is x . Its domain is $\mathbb{R}$. **(a)** Find $\operatorname{arccot}(-1)$ exactly. **(b)** Approximate $\operatorname{arccot}(-2)$ in radians to two decimal places.

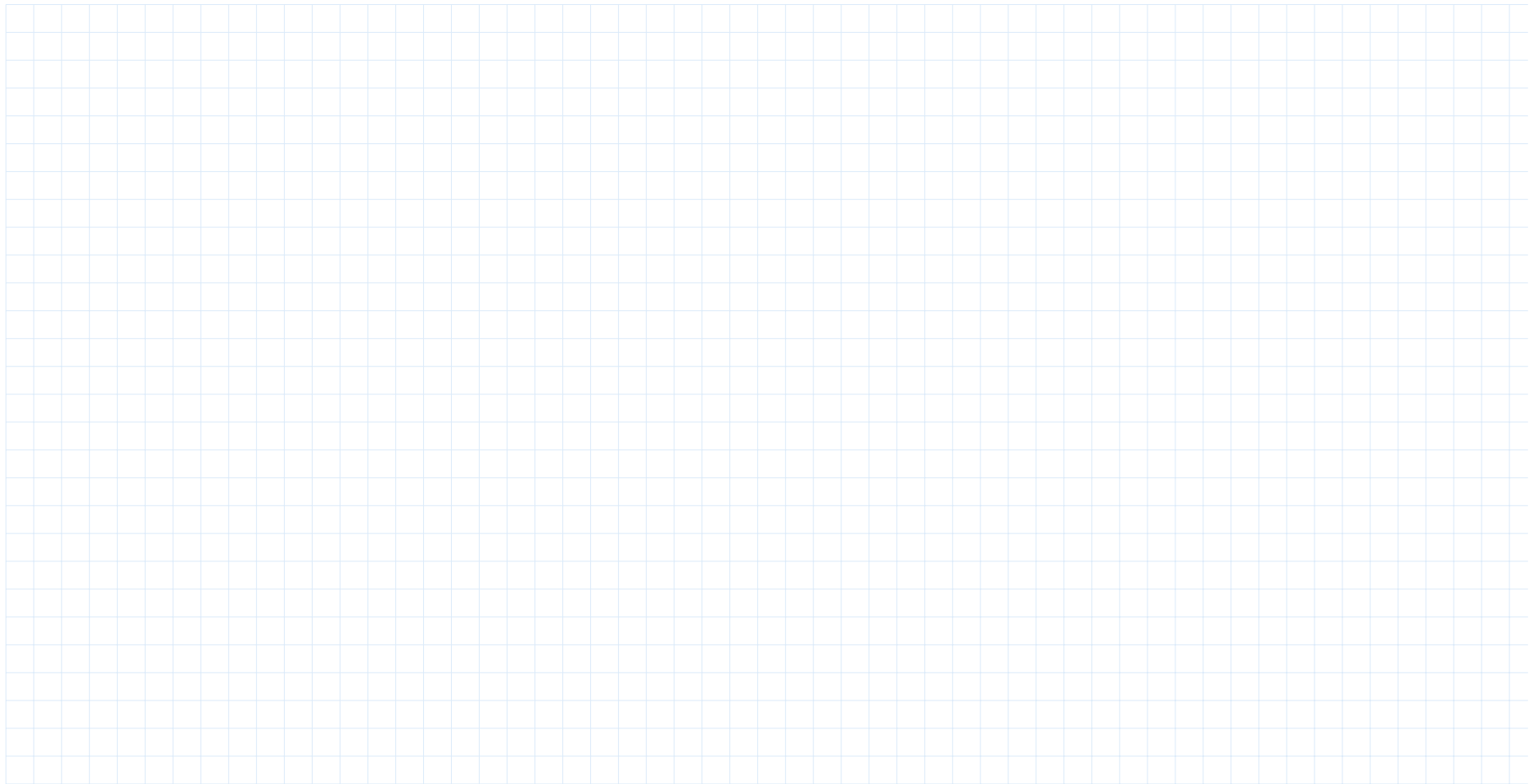


Calculator Conversions

Section 6.2

Rewrite using inverse sine or inverse cosine, then approximate in radians to two decimal places.

(a) $\operatorname{arcsec}(-3)$ (b) $\operatorname{arccsc}(4)$.



Write an Algebraic Expression

Rewrite each expression using only u , arithmetic, and radicals. State the allowed values of u .

(a) $\sin(\arctan u)$

(b) $\cos(\operatorname{arccot} u)$.

